

POLI 110 — Bases of Claims & the Severity Principle

Week 4 Notes

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1 Learning Goals and Overview

This week you will practice moving from *claims* to *tests*. By the end of this week, you should be able to (i) state a claim in a single falsifiable sentence; (ii) identify the *evidence* and the often-implicit *warrant* that links that evidence to the claim; and (iii) judge whether the evidence meets the **Severity Principle**: strong evidence comes from procedures that could have revealed the claim's flaws but did not (*see slide 2–4*). You will then apply this to seven real-world election claims (*see slide 6–12*).

2 Conceptual Foundations

A **claim** is a statement about the world that could be true or false; **evidence** is the information or reasoning that supports it; and a **warrant** links the two by spelling out why the evidence would bear on the claim (*see slide 2*). The **Severity Principle** demands that methods be capable of detecting the claim's falsity if it were false.¹ Four working criteria make severity actionable in introductory research: transparent procedures, consideration of alternatives, systematic use of evidence, and explicit uncertainty (*see slide 4*). Throughout, we distinguish these scientific bases from unscientific ones (authority, personal experience, common sense, emotion/morality, tradition) (*see slide 5*).

Why severity here? A bridge to the reading. This week's article shows why non-severe bases are tempting yet unreliable. In a welfare-policy survey, many respondents held incorrect factual beliefs (e.g., about the share of the federal budget spent on welfare) and were often *highly confident* in those wrong beliefs; confidence and misperception were correlated.² Such "I know I'm right" dynamics help explain why merely asserting credentials or reporting a single dramatic pattern is not strong evidence.³

3 Applying Severity to Election Claims (Slides 6–12)

The table below summarizes each example from slides 6–12 and shows (a) the main non-scientific basis, (b) why it fails the severity test, and (c) what a *severe test* would minimally require.⁴

4 Worked Example

Consider slide 6. The implicit warrant is that winning more counties implies winning more votes. This reasoning fails when population is unevenly distributed. A severe test would replace this warrant with a model that accounts for population: specify a county-level data-generating process for votes

¹Formulation adapted from slide 3; the slides emphasize that agreement between data x and claim C is not probative if the method was practically guaranteed to agree. (*see slide 3*)

²See Kuklinski et al., *The Journal of Politics* 62(3): 790–816, esp. Figure 1 on factual accuracy (p. 797) and Figure 2 on confidence among the misinformed (p. 799).

³Kuklinski et al. also document limited changes in preferences when respondents are simply given correct facts (Table 1, p. 802) and show how misinformation can bias collective preferences (Figure 7, p. 809).

⁴Examples paraphrased from slides 6–12.

Table 1: Scientific vs. Unscientific Bases

Basis	Status	Why it does/doesn't meet severity
Transparent, pre-specified procedure	Scientific	Others can follow the steps; failure modes are visible; falsifiability is clear
Comparative test with plausible alternatives	Scientific	Competing explanations are directly confronted (placebo/negative controls where possible)
Systematic use of data	Scientific	Clear rules for inclusion, measurement, modeling, and inference
Uncertainty acknowledged	Scientific	Reports intervals, assumptions, and sensitivity
Authority/experience/common sense/emotion/tradition	Unscientific	Offers no way the claim could have failed; prone to selective perception and confirmation

Table 2: Slide Exercise: From Claims to Severe Tests

Slide	Claim	Main basis	Why it fails severity	What a severe test would require
6	Record-high votes with record-low counties	Common sense	County counts ignore population clustering; no explicit falsifier	Model vote totals with population/urbanization/turnout; predefine expected frequencies
7	Losing most bellwether counties	Tradition/survivor bias	Post-hoc selection; no out-of-sample performance	Pre-register bellwethers and test predictive accuracy using historical elections
8	Overnight lead reversals are extremely improbable	Personal incredulity	No model of batch arrivals; ignores state counting rules	Simulate timestamped updates by ballot type/locale; compute proper p -values
9	Technical analysis by expert reveals flaws	Authority + jargon	Opaque data/code; no alternatives; no error rates	Release data/code, pre-specify anomaly metrics; replicate on independent data
10	"Go ask anyone in New Hampshire" (bused voters)	Hearsay/common knowledge	No auditable evidence or threshold for materiality	Privacy-preserving linkage of voter files + transportation logs
11	"Everybody knows it was rigged"	Common knowledge + emotion	Not falsifiable; ignores structural shifts	Precinct-level models vs. priors; report uncertainty and sensitivity
12	Midnight ballot "dumps" extremely skewed	Anecdote/single episode	Treats batch reports as random; ignores reporting procedures	Reconstruct batch composition; benchmarks for expected patterns

conditional on population density, urbanization, and turnout; fit the model using historical elections; then compute the expected frequency that a candidate can win the popular vote while carrying relatively few counties. If the observed pattern lies within the model's predictive distribution, the

claim fails; if it is an outlier under plausible models, the claim gains evidential support.

5 Analysis Template for Your Assignments

Step 1: State the claim in one falsifiable sentence

Step 2: Map the basis: What kind of support is being offered (authority, anecdote, etc.) and what is the implicit warrant?

Step 3: Design a severe test: Specify data, a comparison or counterfactual, and a decision rule that could *in principle* overturn the claim

Step 4: Plan alternatives and checks: List rival explanations and at least one placebo/negative-control test

Step 5: Specify uncertainty: What interval or sensitivity analysis will you report, and what result would change your mind?

6 Severity Checklist for Peer Review

Table 3: Use this checklist when reviewing your peers' work

Severity Requirement	Evidence to look for
Transparent procedure	Data sources; inclusion rules; reproducible steps (code or detailed description)
Alternatives confronted	Pre-registered comparisons; placebo/negative controls; robustness to different choices
Systematic use of evidence	Clear measurement definitions; consistent modeling and diagnostics
Uncertainty acknowledged	Standard errors/intervals; sensitivity to assumptions; clearly stated decision thresholds
Outcomes that would falsify	Concrete patterns that, if observed, would count against the claim

7 Connecting to This Week's Reading

The reading's empirical results show why severity is necessary in political argument. Survey respondents frequently misestimated concrete welfare facts, often in the same *direction* (anti-welfare) and with high confidence; confidence and error tended to occur together.⁵ Simply presenting correct facts had limited effect on policy preferences in the short run (Table 1, p. 802), and simulations showed how misinformation can shift *collective* preferences, not just individual views (Figure 7, p. 809). This reinforces a core lesson: because people can feel certain and still be wrong, only tests that could have revealed error—and did not—provide strong evidence.

⁵See Figure 1 on accuracy (p. 797) and Figure 2 on confidence (p. 799).

Additional Reading

Kuklinski, James H., Paul J. Quirk, Jennifer Jerit, David Schwieder, and Robert F. Rich. 2000. "Misinformation and the Currency of Democratic Citizenship." *The Journal of Politics* 62(3): 790–816.